

PRACTICAL WEEK 7

Task 1:

Code:

```
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;

void code_function()
{
    printf("Code address  : %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    *heap_var = 40;

    printf("Memory Addresses\n");
    printf("-----\n");

    code_function();

    printf("Global address : %p\n", (void *)&global_var);

    printf("Static address : %p\n", (void *)&static_var);

    printf("Heap address  : %p\n", (void *)heap_var);

    printf("Stack address : %p\n", (void *)&stack_var);
```

```
    free(heap_var);

    return 0;
}
```

Task 2:

Code:

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    char filename[100];

    printf("Process ID: %d\n", getpid());

    snprintf(filename, sizeof(filename),
              "/proc/%d/maps", getpid());

    printf("\nMemory Mappings\n");
    printf("=====\n");

    FILE *file = fopen(filename, "r");

    if (file == NULL)
    {
        perror("Unable to open /proc maps");
        return 1;
    }

    char line[500];

    while (fgets(line, sizeof(line), file))
    {
        printf("%s", line);
    }

    fclose(file);

    return 0;
}
```

Output:

```
[rishivarma@rishis-MacBook-Air Exp 7 practical % nano memory_addresses.c ]
[rishivarma@rishis-MacBook-Air Exp 7 practical % gcc memory_addresses.c -o memory_addresses ]
[rishivarma@rishis-MacBook-Air Exp 7 practical % ./memory_addresses ]
Memory Addresses
=====
Code address   : 0x1020b04f8
Global address : 0x1020b8000
Static address : 0x1020b8004
Heap address   : 0x102915880
Stack address  : 0x16dd4f018
[rishivarma@rishis-MacBook-Air Exp 7 practical % nano proc_maps.c ]
[rishivarma@rishis-MacBook-Air Exp 7 practical % gcc proc_maps.c -o proc_maps ]
[rishivarma@rishis-MacBook-Air Exp 7 practical % ./proc_maps ]
Process ID: 34536

Memory Mappings
=====
Unable to open /proc maps: No such file or directory
```